# Lecture 13

Corey Powell

2026-10-21

# Lecture 13

1. Random Variables
2. Expected Value
3. Standard Error
4. Covariance

Finish reading Chapters 16-17 for next time.

Quiz 6 is Thursday October 22 in lab.

# Random Variables

# Variables

- Random variables can seem very confusing at first.
- It helps to step back and think about the meaning of variables in mathematics more generally.

# Variable in Mathematics

| Example | Comments |
|---|---|
| $4x + 2 = 10$ | $x$ "unknown," but can we can solve for it. |
| $x^2 = 9$ | $x$ "unknown," but we can partially solve for it. |
| $c = 299,792,458$ | We know $c$. |
| $t = \text{Number of trees}$ | We don't know $t$, but it has a definite value |
| $\sum_{i=1}^{5} i^2$ | $i$ stands for many numbers at once |
| $f(x) = x^2$ | $x$ can be any number we choose |
| $D = \text{the number on a die}$ | $D$ can be one of several possible values. Each value has a probability. |

# Random Variables

- A **random variable** is just a letter, often a capital letter, that stands for the outcome of a chance process.
- A random variable can take on any one of several **possible values**.
- Each of the possible values has a probability.

# Example: Rolling a die

- Suppose you roll a die, and let $X$ denote the number that comes up.
- $X$ is a random variable.
  - The possible values of $X$ are $1, 2, 3, 4, 5, 6$.
  - The probability that $X$ equals 1 is $1/6$.
  - The probability that $X$ equals 2 is $1/6$, and so on.
- Notation: $P(X = 1)$ means the probability that $X$ equals 1.
  - Thus, for example, $P(X = 1) = 1/6$.
- Note: $P(...)$ doesn't represent a function. It just means the probability of.

# Example: Drawing a ticket from a box

- Suppose you draw a ticket at random from a box with seven tickets: $\{2, 5, 5, 7, 7, 7, 9\}$
- You let $X$ denote the number that comes up.
- $X$ is a random variable with possible values $2, 5, 7, 9$.

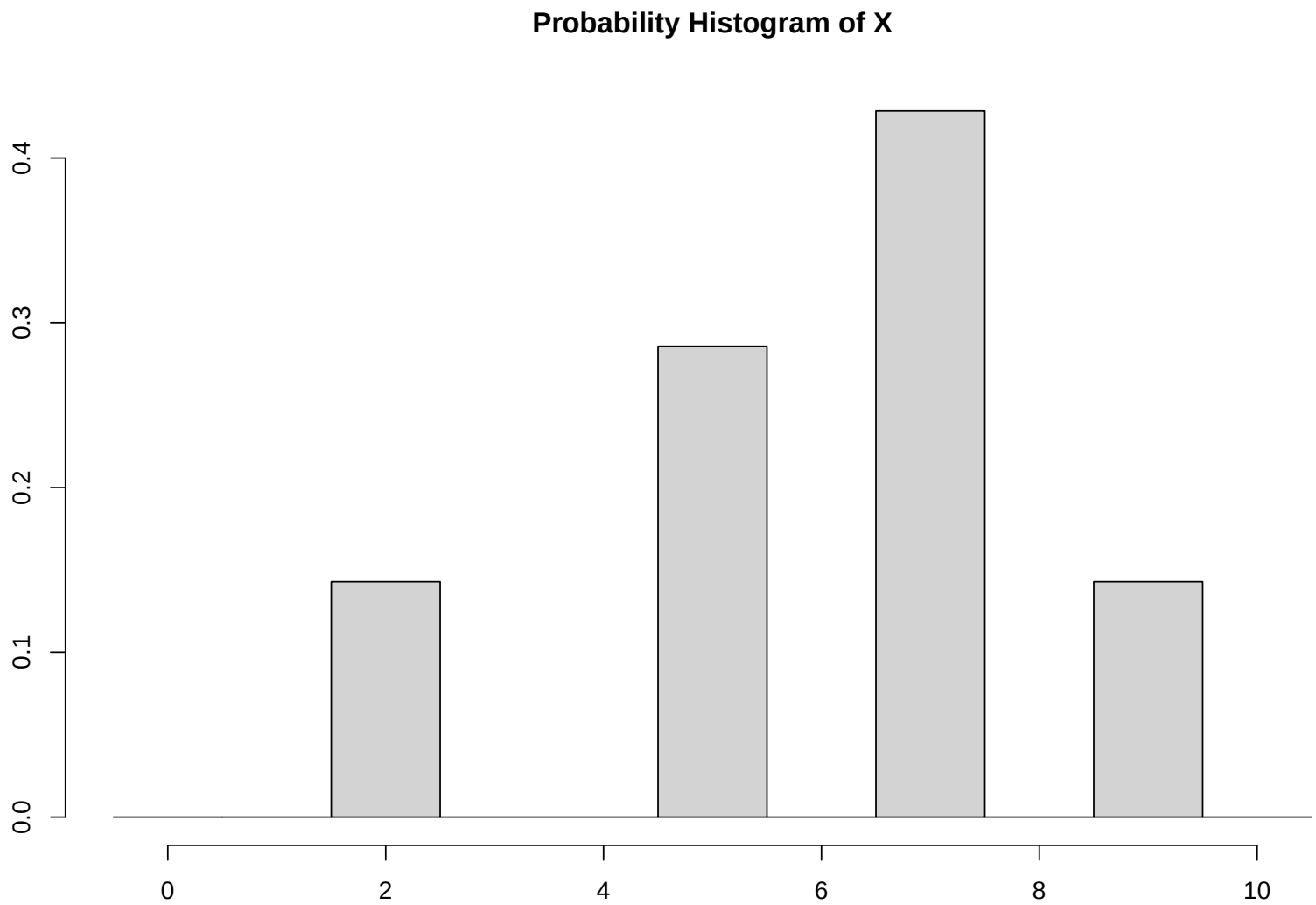
| Possible Value | Probability |
|---|---|
| 2 | 1/7 |
| 5 | 2/7 |
| 7 | 3/7 |
| 9 | 1/7 |

# Probability Histograms

- We can use histograms to represent the distribution of a random variable.
- The area of each bar represents the probability that the random variable will be in a certain range.
- The key difference with data histograms:
  - a probability histogram shows probabilities for a **single random variable**, whereas
  - a data histogram shows fractions of numbers in a list.

# Probability Histograms - Example

- Suppose you draw a ticket at random from a box with seven tickets: $\{2, 5, 5, 7, 7, 7, 9\}$.
- You let $X$ denote the number that comes up.



# R Random Variable Implementation

```
#Creates a sample of n independent random variables
#With distribution from the previous slide
sample_x = function(n) {
  #Create n random numbers between 0 and 1
  p = runif(n)
  #The component of the vector below is 2 if p < 1/7,
  #5 if 1/7 <= p < 3/7, 7 if 3/7 <= p < 6/7, and
  #9 if if p >= 6/7
  return(2*(p<1/7) + 5*(p>= 1/7 & p<3/7) +
         7*(p>=3/7 & p<6/7) + 9*(p>=6/7))
}
```

# Transforming Random Variables

- Functions of random variables are also random variables.
- Example: Suppose a box has tickets $\{1, 2, 2, 3\}$.
  - You randomly draw a ticket, and let the number you draw be $X$.
  - Then $Y = X^2$ is also a random variable.
- What are the possible values of $Y$?
- What are their probabilities?

# Combining Random Variables

- You can combine random variables to create new random variables.
- Example: Suppose a box has tickets $\{1, 2, 2, 3\}$.
  - You randomly draw a ticket, and let the number you draw be $X$.
  - You randomly draw a second ticket, and let the number you draw be $Y$.
  - Then $Z = X + Y$ is also a random variable.
  - So is $XY$ or $(X + 3)Y^2$ or any other function of $X$ and $Y$.
- What is $P(Z = 4)$?

# Combining Random Variables 2

- Answer: Can't say!
- It depends on if we draw with or without replacement.
  - With replacement, $P(Z = 4) = 3/8$.
  - Without replacement, $P(Z = 4) = 1/3$.

# Moral

- When combining random variables to form a new random variable, there is no automatic way to learn the probabilities or possible values of the new random variable.
- It all depends on the relationship between the original random variables.
  - They could be independent, mutually exclusive, or have any other relationship.

# Random Variables Discussion Question

- Suppose a box has tickets $\{1, 2, 2, 3\}$.
- You draw two tickets without replacement.
- Let the first ticket be $X$ and the second ticket be $Y$.
- Let $Z = X(Y - 1)$.
  - What are the possible values of $X, Y, Z$?
  - What is $P(Z = 4)$? $P(Z = 0)$?

# Expected Value

- Let $X$ be a random variable whose possible values are $x_1, x_2, x_3, \ldots, x_n$.
- The **expected value** of $X$ is $E(X) = \sum_{i=1}^{n} x_i P(X = x_i)$.
- This is a weighted average, with the weights being the probabilities.
- Let's think of a list of numbers as a random variable $Y$.
  - Each number in the list has probability $k/n$.
  - The value $k$ is the number of times that the number is in the list.
  - The value $n$ is the number of numbers in the list.
- What would be $E(Y)$?

# Expected Value Example

- Suppose a box has tickets $\{0, 4, 4, 8\}$.
- You draw a ticket and call it $X$.
  - The possible values of $X$ are $0, 4, 8$.
  - So $n = 3$ and $x_1 = 0$, $x_2 = 4$, and $x_3 = 8$.
- The expected value of $X$ is
  $E(X) = 0 * P(X = 0) + 4 * P(X = 4) + 8 * P(X = 8)$.
- $E(X) = 0 * 1/4 + 4 * 2/4 + 8 * 1/4 = 4$.

# Expected Value Interpretation

- $E(X)$ can be thought of as the long-run average of $X$.
- Long-run average is a confusing term!
  - $X$ just denotes a single number.
- Imagine an very, very large number of independent, identical copies of $X$.
  - Take the average of this list.

# Expected Value Interpretation 2

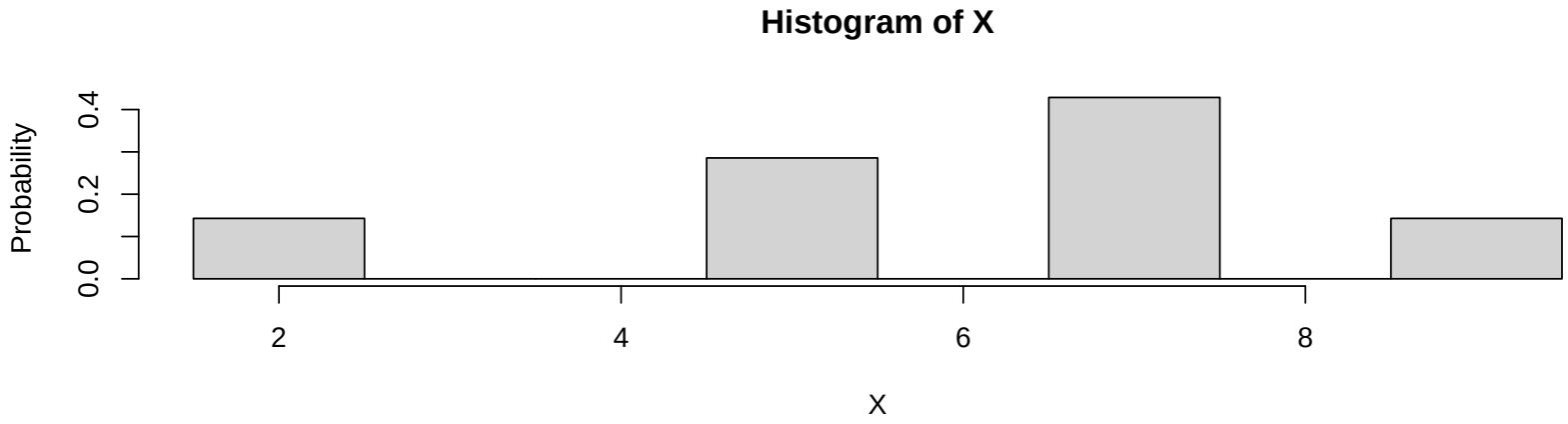
- The expected value is not necessarily expected.
  - For example, the expected value of a die roll is 3.5.
- The expected value is just an ordinary number. It is **not** random.
- The notation $E(X)$ just means the expected value of $X$.
  - It is not a function!

# Expected Value Interpretation Using R

```
X=sample_x(100000000) #Sample of size 100,000,000
mean(X) #Approximately E(X)
```

```
## [1] 6.000138
```

```
hist(X,freq=FALSE,breaks = 2:10-0.5, ylab="Probability")
```



# Useful Formulas for Expected Value

- Suppose $X$ and $Y$ are random variables and that $c$ is some constant value.
- The following formulas are analogous to the formulas for the mean:
  - $E(c) = c$,
  - $E(c + X) = c + E(X)$,
  - $E(cX) = cE(X)$, and
  - $E(X + Y) = E(X) + E(Y)$
- The formula for the sum applies for any $X$ and $Y$.
  - This is more surprising than it might seem at first glance.

# Independent Random Variables

- Two random variables $X$ and $Y$ are **independent** if $P(X = x|Y = y) = P(X = x)$ for any values of $x$ and $y$.
  - The values of $Y$ don't have any influence on the values of $X$, and vice versa.
- This implies that

$$P(X = x \text{ and } Y = y) = P(X = x)P(Y = y)$$

  for any values of $x$ and $y$.
- It follows that $E(XY) = E(X) * E(Y)$.

# Standard Error

- Let $X$ be a random variable.
- The **variance** of $X$ is $\mathrm{Var}(X) = E\left[(X - E(X))^2\right]$.
  - The **standard error** of $X$ is $SE(X) = \sqrt{\mathrm{Var}(X)}$.
- These formulas are analogous to the formulas for the variance and standard deviation of a list of numbers.

# Standard Error Example

- Suppose a box has tickets $\{0, 4, 4, 8\}$.
- You draw a ticket and call it $X$.
- The expected value of $X$ is 4, so $\mathrm{Var}(X) = E\left[(X-4)^2\right]$
- The possible values of $X$ are $0, 4, 8$, so the possible values of $(X-4)^2$ are $0, 16$.
- $P\left[(X-4)^2 = 0\right] = 1/2$ and $P\left[(X-4)^2 = 16\right] = 1/2$
- $E\left[(X-4)^2\right] = 0 \times (1/2) + 16 \times (1/2) = 8$
- Thus $\mathrm{Var}(X) = 8$ and $SE(X) = 2\sqrt{2}$.

# Standard Error Interpretation

- $SE(X)$ can be thought of as the long run standard deviation of $X$.
- The standard error tells us by about how much we might expect $X$ to differ from its expected value.
- The standard error is a measure of the size of the chance variability in $X$.
- The variance is theoretically useful, but is **not** a measure of the size of chance variability.
- The standard error is just an ordinary number. It is *not* random.
- If we were to think of a list of numbers as a random variable $Y$ as before, then what would be $SE(Y)$?

# Standard Error Interpretation Using R

```r
#Simulate 100,000,000 draws for random variable
#from 2 slides ago
p = runif(100000000)
X = 0*(p<1/4) + 4*(p>=1/4 & p<3/4) + 8*(p>=3/4)
#SE(X)
print(sd(X)*sqrt(99999999/100000000))
```

```
## [1] 2.828491
```

```r
#Should be approximately 2*sqrt(2)
print(2*sqrt(2))
```

```
## [1] 2.828427
```

# Covariance

- Let $X$ and $Y$ be a random variables.
- The **covariance** of $X$ and $Y$ is

$$\mathrm{Cov}(X, Y) = E\left[(X - E(X))(Y - E(Y))\right].$$

- The **correlation** of $X$ and $Y$ is $\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X,Y)}{SE(X)SE(Y)}$.
- These formulas are analogous to the formulas for covariance and correlation with a list of numbers.

# Covariance Example 1

- Suppose a box has tickets $0, 4, 4, 8$.
- You draw a ticket without replacement and call it $X$.
- You draw a second ticket and call it $Y$.
- What is the covariance of $X$ and $Y$?

# Table for $X$ and $Y$

| $x$ | $y$ | $P(X = x \text{ and } Y = y)$ |
|---|---|---|
| 0 | 4 | $1/4 * 2/3 = 1/6$ |
| 0 | 8 | $1/4 * 1/3 = 1/12$ |
| 4 | 0 | $1/2 * 1/3 = 1/6$ |
| 4 | 4 | $1/2 * 1/3 = 1/6$ |
| 4 | 8 | $1/2 * 1/3 = 1/6$ |
| 8 | 0 | $1/4 * 1/3 = 1/12$ |
| 8 | 4 | $1/4 * 2/3 = 1/6$ |

# Covariance Example 1 Part 1

- First:
  - What is the expected value of $X$? (Answer: 4.)
  - What is the expected value of $Y$? (Answer: 4.)
- Second:
  - What are the possible values and probabilities of $(X-4)(Y-4)$?
- Third:
  - What is $\mathrm{Cov}(X, Y)$?

# Covariance Example 1 Part 2

| Possible Value | Probability |
|---|---|
| 0 | 5/6 |
| -16 | 1/6 |

- Hence $\mathrm{Cov}(X, Y) = 0 \times (5/6) + (-16) \times (1/6) = -8/3$

# Covariance Example 2

- Suppose a box has tickets $0, 4, 4, 8$.
- You draw a ticket with replacement and call it $X$.
- You then draw a second ticket and call it $Y$.
- What is the covariance of $X$ and $Y$?

# Covariance Example 2 Part 1

- First:
  - What is the expected value of $X$? (Answer: 4)
  - What is the expected value of $Y$? (Answer: 4)
- Second:
  - What are the possible values of $(X-4)(Y-4)$?
  - What are the probabilities?
- Third:
  - What is $\mathrm{Cov}(X, Y)$?

# Covariance Example 2 Part 2

| Possible Value | Probability |
|---|---|
| -16 | 1/8 |
| 0 | 3/4 |
| 16 | 1/8 |

- Hence

$$\mathrm{Cov}(X, Y) = -16 * (1/8) + 0 * (3/4) + 16 * (1/8) = 0.$$

- Is this surprising?

# Useful Formulas For Covariance

- These formulas are analogous for the formulas for covariances and variances with lists of numbers.
- First of all, we have the following shortcut formulas:
  - $\mathrm{Cov}(X, Y) = E(XY) - E(X)E(Y)$
  - $\mathrm{Var}(X) = \mathrm{Cov}(X, X) = E(X^2) - (E(X))^2$
- If $X$ and $Y$ are independent, then $\mathrm{Cov}(X, Y) = 0$.
  - If $\mathrm{Cov}(X, Y) = 0$, however, then $X$ and $Y$ are not necessarily independent.

# More Useful Formulas for Variance

- These are again analogous to formulas for the variance of a list of numbers.
  - $\mathrm{Var}(c + X) = \mathrm{Var}(X)$
  - $\mathrm{Var}(cX) = c^2\mathrm{Var}(X)$
  - $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$
- If $X$ and $Y$ are independent, then

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

- The formula above is very important.

# Useful Formulas for the Standard Error

- These are analogous to formulas for the standard deviation of a list of numbers.
- They follow from taking the square root of the formulas for the variance.
  - $SE(c + X) = SE(X)$
  - $SE(cX) = |c|SE(X)$
  - If $X$ and $Y$ are independent, then $SE(X + Y) = \sqrt{[SE(X)]^2 + [SE(Y)]^2}$.

# Interpretation of Covariance

- $\mathrm{Cov}(X, Y)$ can be thought of as the long-run covariance of $X$ and $Y$.
  - We imagine creating a very large number of $(X, Y)$ pairs.
- $\mathrm{Corr}(X, Y)$ can be thought of as the long-run correlation of $X$ and $Y$.
- If $X$ and $Y$ are independent, then $\mathrm{Cov}(X, Y) = 0$.
- If $\mathrm{Cov}(X, Y) = 0$, then $X$ and $Y$ may **or may not** be independent.

# Covariance Using R

```
#Samples Y for a given value of X
sample_y_x = function(x) {
  p = runif(1)
  if (x==0) {return(4*(p<2/3)+8*(p>=2/3))}
  else if (x==4)
  {return(0*(p<1/3)+4*(p>=1/3 & p<2/3)+8*(p>=2/3))}
  else {return(0*(p<1/3)+4*(p>=1/3)) }
}
```

# Covariance Using R Part 2

```
#Samples X
p = runif(1000000)
X = 0*(p<1/4) + 4*(p>=1/4 & p<3/4) + 8*(p>=3/4)
#Samples Y for each value of X
Y=sapply(X,sample_y_x)
#Cov(X,Y)
print(-8/3)
```

```
## [1] -2.666667
```

```
#Should be approximately Cov(X,Y)
print(cov(X,Y))
```

```
## [1] -2.658997
```

# Covariance using R Part 3

```r
#Y independent of X
p = runif(1000000)
Y = 0*(p<1/4) + 4*(p>=1/4 & p<3/4) + 8*(p>=3/4)
#Should be approximately 0
print(cov(X,Y))
```

```
## [1] -8.579337e-05
```

# Discussion Question 2

- Suppose a box contains 4 tickets, each with two numbers on it, one on the left, and one on the right.
- The four pairs of numbers are: $(0,0), (0,1), (1,0), (1,1)$.
- You randomly draw one ticket.
  - Let $X$ be the number on the right, and $Y$ the number on the left.
- Are $X$ and $Y$ independent?
- What is $\mathrm{Cov}(X, Y)$?

# Discussion Question 3

- Suppose a box contains 4 tickets, each with two numbers on it, one on the left, and one on the right.
- The four pairs of numbers are: $(0, 0), (1, 1), (2, 1), (3, 0)$.
- You randomly draw one ticket.
  - Let $X$ be the number on the right, and $Y$ the number on the left.
- Are $X$ and $Y$ independent?
- What is $\mathrm{Cov}(X, Y)$?

# Review Questions

- See Review_questions13.pdf on Canvas.